Isaac G.L. Daviet, M.Sc.

Biotechnology researcher with hands-on experience in immunology, antibody discovery, and flow cytometry, spanning both experimental and computational biology. Skilled in developing and optimizing assays, workflows, and analysis pipelines across B-cell biology, NGS, and single-cell immune profiling in fast-paced industry and research environments. Demonstrated ability to translate complex biological data into actionable insights through strong experimental design, data analysis, and scientific communication. Supported by an interdisciplinary academic background in biochemistry, history, and biotechnology, to bring clear writing, critical analysis, and cross-disciplinary collaboration skills to applied research.

PROFESSIONAL EXPERIENCE

Imprint Labs, New York City (NY)—*Senior Research Associate*

October 2024 - April 2026

- Conducted R&D on B-cell analysis platforms supporting antibody discovery, immune profiling, and autoimmune disease research.
- Developed, implemented & optimized immunological assays, B-cell antigen tetramer staining, and flow cytometry workflows to improve detection, recovery, specificity, and downstream sequencing performance.
- Built and maintained automated pipelines for processing and QC of single-cell and BCR sequencing data, improving reproducibility and turnaround time.
- Lead efforts in optimizing phylogenetic inference models and development of synthetic BCR sequence validation datasets.
- Collaborated cross-functionally with experimental and computational teams to troubleshoot workflows and improve data quality.

Pelago Biosciences, Stockholm (Sweden)—*Laboratory Technician*

November 2022 - June 2024

- Supported drug discovery research using CETSA to assess protein–ligand interactions in cell-based assays.
- Maintained and prepared mammalian cell lines to support assay execution and reproducibility.
- Assisted with assay setup, data collection, and documentation in an industry research environment.
- Contributed to implementation of an electronic lab notebook (ELN) system in collaboration with IT and lab staff.

Antibody Design Labs, San Diego (CA)—*Research Associate II*

July 2020 - July 2022

- Conducted antibody discovery & engineering through phage display screening, recombinant expression, purification, bioconjugation, characterization and assay development.
- Spearheaded development & production of novel phage, plasmid vector & antibody product lines
- Supported multiple antibody discovery campaigns through assay execution, QC, and data interpretation.
- Developed SOPs and mentored junior researchers, supporting team scalability and experimental consistency.

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Github

LANGUAGES & INTERNATIONAL

- English (Fluent)
- French (Fluent)
- Spanish (Conversational)
- USA, EU and Australia work eligibility

SKILLS

Immunology & Wet Lab

- Flow cytometry / FACS (panel design, optimization, FlowJo & in depth analysis)
- Immunological & Biochemical assays (ELISA, western blot, conjugation...)
- Mammalian cell culture & Expression
- Protein and antibody cloning, expression, purification (HPLC/FPLC), and characterization (SDS PAGE, CryoEM & X-Ray crystallography)
- Phage display and library construction
- Molecular biology and DNA techniques
- Bioconjugation
- Liquid chromatography (SEC, Affinity, AKTA systems)

NGS & Single-Cell

- Illumina NGS Sequencing
- Single-cell and BCR sequencing workflows
- Immune repertoire analysis
- NGS data QC and processing

Earlier Roles

2018-19: Schief Lab, Scripps Research Institute (CA) — *Lab Technician* (HIV vaccine research)
2018: Aurora Biolabs (CA) — *Sales Representative* (customer relations, marketing, advert design)
2018: Unlimited Learning (CA) — *Tutor* (Biology & History)
2016-18: Corbett Lab, UCSD (CA) — *Lab Technician* (protein crystallography, molecular biology)
2015-16: Sunahara Lab, UCSD (CA) — *Lab Assistant* (protein crystallization, proteomics)

EDUCATION

M.Sc. of Molecular Techniques in Life Sciences

Karolinska Institutet, KTH Royal Institute of Technology & Stockholm University

August 2022 - June 2024

Interdisciplinary graduate program integrating immunology, bioengineering, bioinformatics, and molecular biology.

- Master's thesis: Applied machine learning and dimensionality reduction to immune repertoire datasets to identify structural and functional patterns relevant to antibody discovery.
- Training across a variety of cutting edge experimental, computational, and engineering approaches to biomedical research, including a 6 month CryoEm research project at the Cryo-EM Swedish National Facility.

B.S. of Biochemistry/B.A. of History (Minor in Archaeology)

University of California - San Diego

September 2014 - December 2018

Dual degree combining rigorous biochemical training with advanced research and writing in the humanities.

- History B.A. awarded with departmental honors and distinction
- Coursework emphasized critical analysis, biochemical techniques, writing, and research methodologies
- Selected coursework at the Rady School of Management in business and management fundamentals

Additional Academic Experience

Institute for Field Research, Andahuaylas (Peru) — *Sondor Bioarchaeology Fieldwork*
University of the Aegean, Phocis (Greece) — *Kastrouli Archaeology Fieldwork*
University of California-Los Angeles, Los Angeles — *Sci/Art Lab Program*

ACADEMIC PROJECTS

- **Master's Thesis (Univ. of Oslo):** *Structural Data Recovery Through Machine Learning Integrated Gradient Analysis* — Applied AI and dimensionality reduction methods to immunological data, expanding potential for antibody discovery & analysis.
- **Cryo-EM Internship (SciLifeLab, Sweden):** Hands-on cryo-electron microscopy project using high-throughput Chameleon System to optimize for Preferential Particle Orientation.
- **History Honors Thesis (UCSD):** *Asgard's Shame; Christianity's Tool; Societal Roles, Seidr Magic and the Paradox of Odin*
- **Staff writer (Under the Scope Magazine):** Science Communication publication adapting complex research into accessible articles, focusing on phage research..
- **Article Contributor (Buck Institute for Research on Aging):** Velarde, M. C., Demaria, M., Melov, S., & Campisi, J. (2015). Pleiotropic age-dependent effects of mitochondrial dysfunction on epidermal stem cells. Proceedings of the National Academy of Sciences, 112(33), 10407–10412. https://doi.org/10.1073/pnas.1505675112
- **Vice President/Founding Member (Ohlone College Biotechnology Club)**

Computational & Data Analysis
- Bioinformatics (Python, R, Bash, Git)
- Data visualization and dimensionality reduction
- Biostatistics and applied phylogenetics
- Synthetic antibody dataset modeling
- Cloud-based workflows
- Structural/Protein modeling (AlphFold, Pymol)

Research & Analysis
- Data analysis & interpretation
- QC Documentation
- Literature review
- Critical analysis

Management & Organization
- Lab management
- Database management
- Project coordination
- Quality Assurance & Control

Communication
- Scientific writing, communication & presentation
- SOP development
- Staff training
- Interdisciplinary collaboration